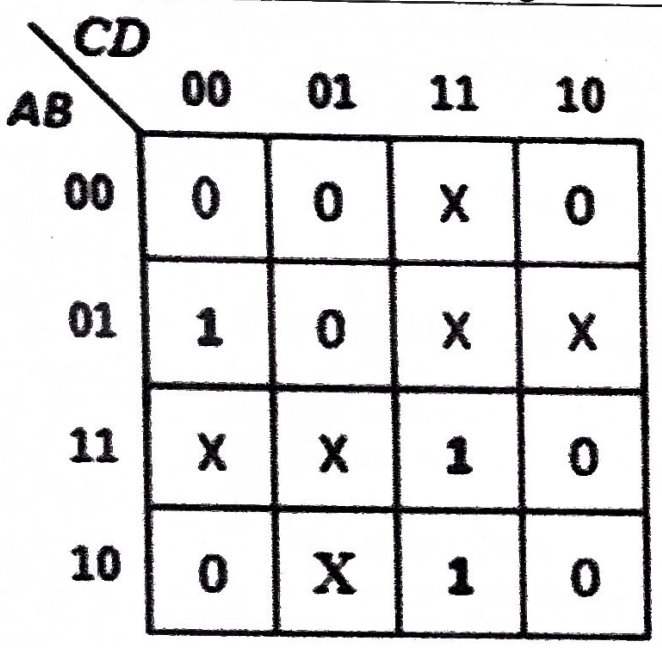
Academic year 2025-2026 (Odd Sem)

DEPARTMENT OF

COMPUTER SCIENCE & ENGINEERING

Date	Nov 2025	Maximum Marks	60
Course Code	CS234AI	Duration	120 Min
Sem	III Semester	Test-1	
ADLD & CO			

Part A

Sl. No.	Questions	M	BT	CO
1.1	Represent −5.75in IEEE 754 single precision format.	2	1	1
1.2	Give the truth table and circuit diagram for a 1 bit comparator.	2	3	3
1.3	 Find out the number of prime and essential prime implicants for the above K-map	2	2	2
1.4	Write the booth recoding for multiplier (binary) for 11011001101	1	3	3
1.5	Implement the function F(A,B)=A+B′ using a 2-to-1 MUX.	1	1	4
1.6	Implement the Boolean function: $F(A,B,C) = \prod M(0,2,4,5)$ using a **3-to-8 line maxterm generator.**	1	2	2
1.7	Perform: (−27)−(+14)using 2's complement (8-bit).	1	3	2

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2. a)	Use **bit-pair recoding** to multiply (-13) × 11.	4	1	1
b)	Give the flow chart for restoring division and perform the division operation on following number. Dividend: 8 Divisor: 5	6	1	1
3. a)	Design a three input, one output minimal two-level gate combinational network that has a logic 1 output when the majority of its inputs are logic 1 and has a logic 0 output when the majority of its inputs are logic 0.	5	2	2
b)	Using the Quine-McCluskey method, obtain all the prime implicants for the following Boolean functions. $F(w,x,y,z) = \sum m(0,2,3,4,8,10,12,13,14)$	5	3	3
4. a)	Design a digital circuit using a 4-bit binary adder to convert a decimal digit represented in 8421 BCD code into its corresponding Excess-3 code. (a) Construct the truth table showing the relationship between the 8421 BCD inputs and the Excess-3 outputs. (b) Explain the design approach used for the conversion. (c) Verify your design using the example: **A = 1001** and **B = 0011**. Perform the addition and confirm that the output represents the correct Excess-3 code.	5	4	4
b)	Design and implement a 4-bit Carry Look-Ahead (CLA) Adder. Show the logic diagram, derive the carry and sum expressions, and explain how carry propagation is improved compared to a ripple-carry adder.	5	2	2
5. a)	Design a binary-to-Gray code converter using a 4-to-16 line decoder constructed with NAND gates and equipped with an enable input (E'). (a) Draw or describe the design approach and connections used to obtain the Gray code outputs. (b) Prepare the truth table showing the relationship between the 4-bit binary inputs and the corresponding 4-bit Gray code outputs. (c) Derive the logic expressions for each Gray code output in terms of the binary inputs. (d) Verify the correctness of your design with a suitable example.	5	1	1
5.b	Design a decoder-based alarm circuit that triggers an alarm if a 4-bit sensor input equals 1100, 1110, or 1011. Use a 4-to-16 decoder and an OR gate to combine the required outputs.	5	3	4
6.a	Design a multiplexer tree to implement a 16-to-1 line multiplexer using a combination of 2-to-1 and 8-to-1 multiplexers. (a) Draw or describe the circuit showing all inputs, select lines, and output connections clearly. (b) Explain the working principle of the multiplexer tree. (c) Verify your design with a suitable example input selection, showing how the correct data input is routed to the output.	4	1	1
6.b	Realize the Boolean expression $F(w,x,y,z) = \sum m(4,5,7,8,10,12,15)$ using a 4-to-1 line MUX and external gates. Let y and z appear on select line S1 and S0 respectively.	6	3	4